

МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ РОССИЙСКОЙ
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«ВЯТСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ»

Институт математики и информационных систем
Факультет автоматики и вычислительной техники
Кафедра электронных вычислительных машин

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Разработка приложения Formula Racing Management System на Java с
использованием Spring Boot и PostgreSQL.

Отчёт по лабораторной работе №3
по дисциплине

«Объектно-ориентированное программирование»

Разработал студент гр. ИВТб-2301-05-00

_____/Черкасов А. А./
(подпись)

Преподаватель

_____/Шмакова Н. А./
(подпись)

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Киров
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Цели лабораторной работы

- освоить разработку веб-приложений на языке Java с использованием фреймворка Spring Boot;
- изучить применение ORM-инструмента Hibernate для работы с реляционной базой данных PostgreSQL без написания SQL-запросов вручную;
- реализовать полноценное CRUD-приложение с веб-интерфейсом и нормализованной схемой БД;
- изучить инструмент управления миграциями базы данных Flyway.

Задание

Разработать веб-приложение «Formula Racing Management System» на языке Java, соответствующее следующим требованиям:

1. Использовать язык программирования Java.
2. Использовать СУБД PostgreSQL.
3. Использовать сборщик Maven (зависимости только через `pom.xml`).
4. Использовать Hibernate для работы с сущностями без SQL-запросов.
5. Нормализованная схема БД: не менее трёх связанных таблиц, структура соответствует ЗНФ.
6. Реализовать CRUD-операции: добавление, просмотр, редактирование и удаление объектов.
7. Пользовательский интерфейс реализован в виде HTML-страниц с использованием шаблонизатора Thymeleaf.

Реализация приложения

Предметная область

Система автоматизирует управление гоночными командами в трёх чемпионатах: Formula 1, Formula 2 и Formula E. Каждая серия объединяет несколько команд, а каждая команда включает пилотов с личными очками. Иерархия объектов отражает реальные данные сезона 2026 года.

Архитектура приложения

Проект построен по принципу многоуровневой архитектуры (Layered Architecture) в рамках Spring Boot:

- **Модели (Models)** — сущности JPA, отражающие таблицы БД: `RacingSeries`, `RacingTeam`, `RacingDriver`.
- **Репозитории (Repositories)** — интерфейсы Spring Data JPA, автоматически предоставляющие CRUD-операции и производные запросы (например, `findByTeam_SeriesIdOrderByPointsDesc`).
- **Сервис (Service)** — слой `RacingService`, инкапсулирующий бизнес-логику и изолирующий контроллер от деталей работы с данными.
- **Контроллер (Controller)** — `RacingController`, обрабатывающий HTTP-запросы и передающий данные в шаблоны Thymeleaf.
- **Представление (View)** — HTML-шаблоны с разметкой Thymeleaf (`index.html`, `team-form.html`, `driver-form.html`).

Схема базы данных

Для хранения данных используется PostgreSQL, развёрнутая в Docker-контейнере. Схема используется лабораторными работами №2, №3 и №4.

Схема нормализована до третьей нормальной формы (3НФ) и содержит три основные таблицы:

- **racing_series**: хранит информацию о чемпионатах (название, официальное наименование, штаб-квартира, управляющая организация, год основания, количество этапов).
- **racing_teams**: хранит данные о командах (название, руководитель, база, силовая установка, очки). Таблица расширена дополнительными полями (победы, подиумы, бюджет и др.) для поддержки специфических полей GUI-приложения из ЛР №2. Каждая команда ссылается на серию через внешний ключ `series_id`.
- **racing_drivers**: хранит данные о пилотах (имя, фамилия, гоночный номер, очки). Каждый пилот привязан к команде через `team_id`.

Управление миграциями осуществляется через **Flyway**: при запуске приложения автоматически применяются миграции V1–V5. Скрипты создают таблицы, наполняют их реальными данными сезона 2026 года и расширяют структуру под потребности дочерних классов C#-приложения.

JPA-сущности и связи

Связи между сущностями реализованы через аннотации JPA:

```
// B RacingSeries.java
@OneToMany(mappedBy = "series", cascade = CascadeType.ALL)
private List<RacingTeam> teams;

// B RacingTeam.java
@ManyToOne(fetch = FetchType.LAZY)
@JoinColumn(name = "series_id", nullable = false)
private RacingSeries series;

@OneToMany(mappedBy = "team", cascade = CascadeType.ALL)
private List<RacingDriver> drivers;
```

Такой подход позволяет Hibernate автоматически выстраивать граф объектов при загрузке.

Оптимизация запросов: N+1 и @EntityGraph

При отображении личного зачёта пилотов по сериям возникает проблема N+1 запроса: после загрузки списка пилотов Hibernate выполняет отдельный SQL-запрос для каждого связанного объекта `team` и `team.series`. Для решения используется `@EntityGraph` на производном запросе Spring Data:

```
public interface RacingDriverRepository extends JpaRepository<RacingDriver, Long> {  
    @EntityGraph(attributePaths = {"team", "team.series"})  
    List<RacingDriver> findByTeam_SeriesIdOrderByPointsDesc(Long seriesId);  
}
```

Аннотация `@EntityGraph` указывает Hibernate загрузить связанные сущности одним LEFT JOIN запросом (используя JOIN FETCH), полностью устраняя N+1. Важно: `@EntityGraph` применяется только к конкретному методу, а не к `findAll()`, чтобы избежать ненужных JOIN-ов при простом перечислении всех пилотов.

Сервис формирует `LinkedHashMap<RacingSeries, List<RacingDriver>`», который Thymeleaf итерирует для отрисовки таблицы личного зачёта с разделителями по сериям.

Веб-интерфейс

Интерфейс реализован как многостраничное веб-приложение (7 шаблонов Thymeleaf)

Страницы приложения:

- `index.html` — дашборд с таблицей команд, пилотов и личным зачётом, сгруппированными по сериям;
- `series.html` — просмотр/управление чемпионатами;
- `teams.html` / `drivers.html` — списки команд и пилотов с фильтрацией;
- `series-form.html` / `team-form.html` / `driver-form.html` — формы добавления/редактирования.

Вывод

В ходе выполнения лабораторной работы было разработано веб-приложение на Java с использованием Spring Boot, PostgreSQL, Hibernate и Thymeleaf. Реализованы нормализованная схема БД в 3НФ, миграции Flyway, CRUD-операции для трёх сущностей, группировка данных по сериям и личный зачёт пилотов.

Приложение A1. Сущность RacingSeries.java

```
package com.vyatsu.racing.models;

import javax.persistence.*;
import java.util.List;

@Entity
@Table(name = "racing_series")
public class RacingSeries {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false)
    private String name;

    @Column(name = "official_name")
    private String officialName;

    @Column(nullable = false)
    private String headquarters;

    @Column(name = "governing_body")
    private String governingBody;

    @Column(name = "founded_year")
    private Integer foundedYear;

    @Column(name = "rounds")
    private Integer rounds;

    @OneToMany(mappedBy = "series", cascade = CascadeType.ALL)
    private List<RacingTeam> teams;

    public Long getId() { return id; }
```

```

public void setId(Long id) { this.id = id; }

public String getName() { return name; }
public void setName(String name) { this.name = name; }

public String getOfficialName() { return officialName; }
public void setOfficialName(String officialName) { this.officialName =
↪ officialName; }

public String getHeadquarters() { return headquarters; }
public void setHeadquarters(String headquarters) { this.headquarters =
↪ headquarters; }

public String getGoverningBody() { return governingBody; }
public void setGoverningBody(String governingBody) { this.governingBody =
↪ governingBody; }

public Integer getFoundedYear() { return foundedYear; }
public void setFoundedYear(Integer foundedYear) { this.foundedYear = foundedYear;
↪ }

public Integer getRounds() { return rounds; }
public void setRounds(Integer rounds) { this.rounds = rounds; }

public List<RacingTeam> getTeams() { return teams; }
public void setTeams(List<RacingTeam> teams) { this.teams = teams; }
}

```

Приложение А2. Сущность RacingTeam.java

```

package com.vyatsu.racing.models;

import lombok.Data;
import lombok.ToString;
import javax.persistence.*;
import java.util.List;

```



```

@Entity
@Table(name = "racing_teams")
public class RacingTeam {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false)
    private String name;

    @Column(name = "principal_name", nullable = false)
    private String principalName;

    @Column(name = "base_location", nullable = false)
    private String baseLocation;

    @Column(name = "power_unit")
    private String powerUnit;

    @Column(nullable = false)
    private Integer points = 0;

    @ManyToOne(fetch = FetchType.LAZY)
    @JoinColumn(name = "series_id", nullable = false)
    @ToString.Exclude
    private RacingSeries series;

    @OneToMany(mappedBy = "team", cascade = CascadeType.ALL)
    @ToString.Exclude
    private List<RacingDriver> drivers;

    @Column(name = "team_color")
    private String teamColor = "#FFFFFF";

    // Lab3 V5: extra fields from lab2
    @Column(nullable = false)

```

```

private Integer wins = 0;

@Column(nullable = false)
private Integer podiums = 0;

@Column(name = "budget_cap")
private Double budgetCap = 0.0;

@Column(name = "constructor_pos")
private Integer constructorPos = 0;

@Column(name = "chassis_model")
private String chassisModel;

@Column(nullable = false)
private Integer graduates = 0;

@Column(name = "is_feeder")
private Boolean isFeeder = false;

@Column(name = "energy_partner")
private String energyPartner;

@Column(name = "battery_kwh")
private Double batteryKwh = 0.0;

@Column(name = "sustain_score")
private Integer sustainScore = 0;

public Long getId() { return id; }
public void setId(Long id) { this.id = id; }

public String getName() { return name; }
public void setName(String name) { this.name = name; }

public String getPrincipalName() { return principalName; }

```

```

public void setPrincipalName(String principalName) { this.principalName =
↪ principalName; }

public String getBaseLocation() { return baseLocation; }
public void setBaseLocation(String baseLocation) { this.baseLocation =
↪ baseLocation; }

public String getPowerUnit() { return powerUnit; }
public void setPowerUnit(String powerUnit) { this.powerUnit = powerUnit; }

public Integer getPoints() { return points; }
public void setPoints(Integer points) { this.points = points; }

public String getTeamColor() { return teamColor; }
public void setTeamColor(String teamColor) { this.teamColor = teamColor; }

public RacingSeries getSeries() { return series; }
public void setSeries(RacingSeries series) { this.series = series; }

public List<RacingDriver> getDrivers() { return drivers; }
public void setDrivers(List<RacingDriver> drivers) { this.drivers = drivers; }

public Integer getWins() { return wins; }
public void setWins(Integer wins) { this.wins = wins; }

public Integer getPodiums() { return podiums; }
public void setPodiums(Integer podiums) { this.podiums = podiums; }

public Double getBudgetCap() { return budgetCap; }
public void setBudgetCap(Double budgetCap) { this.budgetCap = budgetCap; }

public Integer getConstructorPos() { return constructorPos; }
public void setConstructorPos(Integer constructorPos) { this.constructorPos =
↪ constructorPos; }

public String getChassisModel() { return chassisModel; }

```

```

public void setChassisModel(String chassisModel) { this.chassisModel =
↳ chassisModel; }

public Integer getGraduates() { return graduates; }
public void setGraduates(Integer graduates) { this.graduates = graduates; }

public Boolean getIsFeeder() { return isFeeder; }
public void setIsFeeder(Boolean isFeeder) { this.isFeeder = isFeeder; }

public String getEnergyPartner() { return energyPartner; }
public void setEnergyPartner(String energyPartner) { this.energyPartner =
↳ energyPartner; }

public Double getBatteryKwh() { return batteryKwh; }
public void setBatteryKwh(Double batteryKwh) { this.batteryKwh = batteryKwh; }

public Integer getSustainScore() { return sustainScore; }
public void setSustainScore(Integer sustainScore) { this.sustainScore =
↳ sustainScore; }
}

```

Приложение А3. Сущность RacingDriver.java

```

package com.vyatsu.racing.models;

import lombok.Data;
import lombok.ToString;
import javax.persistence.*;

@Entity
@Table(name = "racing_drivers")
public class RacingDriver {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

```

```

@Column(name = "first_name", nullable = false)
private String firstName;

@Column(name = "last_name", nullable = false)
private String lastName;

@Column(name = "racing_number", nullable = false)
private Integer racingNumber;

@Column(nullable = false)
private Integer points = 0;

@ManyToOne(fetch = FetchType.LAZY)
@JoinColumn(name = "team_id", nullable = false)
@ToString.Exclude
private RacingTeam team;

public Long getId() { return id; }
public void setId(Long id) { this.id = id; }

public String getFirstName() { return firstName; }
public void setFirstName(String firstName) { this.firstName = firstName; }

public String getLastName() { return lastName; }
public void setLastName(String lastName) { this.lastName = lastName; }

public Integer getRacingNumber() { return racingNumber; }
public void setRacingNumber(Integer racingNumber) { this.racingNumber =
↪ racingNumber; }

public Integer getPoints() { return points; }
public void setPoints(Integer points) { this.points = points; }

public RacingTeam getTeam() { return team; }
public void setTeam(RacingTeam team) { this.team = team; }
}

```

Приложение A4. RacingDriverRepository.java

```
package com.vyatsu.racing.repositories;

import com.vyatsu.racing.models.RacingDriver;
import org.springframework.data.jpa.repository.EntityGraph;
import org.springframework.data.repository.CrudRepository;
import org.springframework.stereotype.Repository;
import java.util.List;

@Repository
public interface RacingDriverRepository extends CrudRepository<RacingDriver, Long> {
    List<RacingDriver> findAllOrderByPointsDesc();

    @EntityGraph(attributePaths = {"team", "team.series"})
    List<RacingDriver> findByTeam_SeriesIdOrderByPointsDesc(Long seriesId);
}
```

Приложение A5. RacingService.java

```
package com.vyatsu.racing.services;

import com.vyatsu.racing.models.RacingDriver;
import com.vyatsu.racing.models.RacingSeries;
import com.vyatsu.racing.models.RacingTeam;
import com.vyatsu.racing.repositories.RacingDriverRepository;
import com.vyatsu.racing.repositories.RacingSeriesRepository;
import com.vyatsu.racing.repositories.RacingTeamRepository;
import org.springframework.stereotype.Service;
import java.util.ArrayList;
import java.util.LinkedHashMap;
import java.util.List;

@Service
public class RacingService {

    private final RacingSeriesRepository seriesRepository;
```

```

private final RacingTeamRepository teamRepository;
private final RacingDriverRepository driverRepository;

public RacingService(RacingSeriesRepository seriesRepository,
                    RacingTeamRepository teamRepository,
                    RacingDriverRepository driverRepository) {
    this.seriesRepository = seriesRepository;
    this.teamRepository = teamRepository;
    this.driverRepository = driverRepository;
}

// Series
public Iterable<RacingSeries> getAllSeries() {
    return seriesRepository.findAll();
}

public List<RacingSeries> getAllSeriesWithTeams() {
    List<RacingSeries> all = new ArrayList<>();
    seriesRepository.findAll().forEach(all::add);
    return all;
}

public RacingSeries getSeriesById(Long id) {
    return seriesRepository.findById(id).orElse(null);
}

public void saveSeries(RacingSeries series) {
    seriesRepository.save(series);
}

public void deleteSeries(Long id) {
    seriesRepository.deleteById(id);
}

// Teams
public Iterable<RacingTeam> getAllTeams() {
    return teamRepository.findAll();
}

```

```

}

public RacingTeam getTeamById(Long id) {
    return teamRepository.findById(id).orElse(null);
}

public void saveTeam(RacingTeam team) {
    teamRepository.save(team);
}

public void deleteTeam(Long id) {
    teamRepository.deleteById(id);
}

// Drivers
public Iterable<RacingDriver> getAllDrivers() {
    return driverRepository.findAll();
}

public List<RacingDriver> getDriverStandings() {
    return driverRepository.findAllOrderByPointsDesc();
}

public LinkedHashMap<RacingSeries, List<RacingDriver>> getStandingsBySeries() {
    LinkedHashMap<RacingSeries, List<RacingDriver>> map = new LinkedHashMap<>();
    List<RacingSeries> series = new ArrayList<>();
    seriesRepository.findAll().forEach(series::add);
    for (RacingSeries s : series) {
        map.put(s,
            ↪ driverRepository.findByTeam_SeriesIdOrderByPointsDesc(s.getId()));
    }
    return map;
}

public RacingDriver getDriverById(Long id) {
    return driverRepository.findById(id).orElse(null);
}

```



```

    public void saveDriver(RacingDriver driver) {
        driverRepository.save(driver);
    }

    public void deleteDriver(Long id) {
        driverRepository.deleteById(id);
    }
}

```

Приложение А6. RacingController.java

```

package com.vyatsu.racing.controllers;

import com.vyatsu.racing.models.RacingDriver;
import com.vyatsu.racing.models.RacingSeries;
import com.vyatsu.racing.models.RacingTeam;
import com.vyatsu.racing.services.RacingService;
import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.*;

@Controller
public class RacingController {

    private final RacingService racingService;

    public RacingController(RacingService racingService) {
        this.racingService = racingService;
    }

    @GetMapping("/")
    public String index(Model model) {
        model.addAttribute("teams", racingService.getAllTeams());
        model.addAttribute("drivers", racingService.getAllDrivers());
        model.addAttribute("seriesList", racingService.getAllSeriesWithTeams());
    }
}

```

```

        model.addAttribute("standingsBySeries",
            ↪ racingService.getStandingsBySeries());
        return "index";
    }

    // --- SERIES ---
    @GetMapping("/series")
    public String viewSeries(Model model) {
        model.addAttribute("seriesList", racingService.getAllSeriesWithTeams());
        return "series";
    }

    @GetMapping("/series/add")
    public String showAddSeriesForm(Model model) {
        model.addAttribute("series", new RacingSeries());
        return "series-form";
    }

    @GetMapping("/series/edit/{id}")
    public String showEditSeriesForm(@PathVariable Long id, Model model) {
        model.addAttribute("series", racingService.getSeriesById(id));
        return "series-form";
    }

    @PostMapping("/series/save")
    public String saveSeries(@ModelAttribute RacingSeries series) {
        racingService.saveSeries(series);
        return "redirect:/series";
    }

    @GetMapping("/series/delete/{id}")
    public String deleteSeries(@PathVariable Long id) {
        racingService.deleteSeries(id);
        return "redirect:/series";
    }

    // --- TEAMS ---

```

```

@GetMapping("/teams")
public String viewTeams(Model model) {
    model.addAttribute("seriesList", racingService.getAllSeriesWithTeams());
    return "teams";
}

@GetMapping("/teams/add")
public String showAddTeamForm(Model model) {
    model.addAttribute("team", new RacingTeam());
    model.addAttribute("seriesList", racingService.getAllSeries());
    return "team-form";
}

@GetMapping("/teams/edit/{id}")
public String showEditTeamForm(@PathVariable Long id, Model model) {
    model.addAttribute("team", racingService.getTeamById(id));
    model.addAttribute("seriesList", racingService.getAllSeries());
    return "team-form";
}

@PostMapping("/teams/save")
public String saveTeam(@ModelAttribute RacingTeam team) {
    racingService.saveTeam(team);
    return "redirect:/teams";
}

@GetMapping("/teams/delete/{id}")
public String deleteTeam(@PathVariable Long id) {
    racingService.deleteTeam(id);
    return "redirect:/teams";
}

// --- DRIVERS ---
@GetMapping("/drivers")
public String viewDrivers(Model model) {
    model.addAttribute("seriesList", racingService.getAllSeriesWithTeams());

```

```

        model.addAttribute("standingsBySeries",
            ↪ racingService.getStandingsBySeries());
        return "drivers";
    }

    @GetMapping("/drivers/add")
    public String showAddDriverForm(Model model) {
        model.addAttribute("driver", new RacingDriver());
        model.addAttribute("teams", racingService.getAllTeams());
        return "driver-form";
    }

    @GetMapping("/drivers/edit/{id}")
    public String showEditDriverForm(@PathVariable Long id, Model model) {
        model.addAttribute("driver", racingService.getDriverById(id));
        model.addAttribute("teams", racingService.getAllTeams());
        return "driver-form";
    }

    @PostMapping("/drivers/save")
    public String saveDriver(@ModelAttribute RacingDriver driver) {
        racingService.saveDriver(driver);
        return "redirect:/drivers";
    }

    @GetMapping("/drivers/delete/{id}")
    public String deleteDriver(@PathVariable Long id) {
        racingService.deleteDriver(id);
        return "redirect:/drivers";
    }
}

```

Приложение А7. Скрипт миграции V1__init.sql

```

CREATE TABLE racing_series (
    id BIGSERIAL PRIMARY KEY,
    name VARCHAR(255) NOT NULL,

```

```

    official_name VARCHAR(512),
    headquarters VARCHAR(255) NOT NULL,
    governing_body VARCHAR(255),
    founded_year INTEGER,
    rounds INTEGER
);

```

```

CREATE TABLE racing_teams (
    id BIGSERIAL PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    principal_name VARCHAR(255) NOT NULL,
    base_location VARCHAR(255) NOT NULL,
    power_unit VARCHAR(255),
    points INTEGER NOT NULL DEFAULT 0,
    series_id BIGINT NOT NULL REFERENCES racing_series(id)
);

```

```

CREATE TABLE racing_drivers (
    id BIGSERIAL PRIMARY KEY,
    first_name VARCHAR(255) NOT NULL,
    last_name VARCHAR(255) NOT NULL,
    racing_number INTEGER NOT NULL,
    points INTEGER NOT NULL DEFAULT 0,
    team_id BIGINT NOT NULL REFERENCES racing_teams(id)
);

```

-- Series

```

INSERT INTO racing_series (name, official_name, headquarters, governing_body,
↪ founded_year, rounds) VALUES
('Formula 1',
 'FIA Formula One World Championship',
 'Paris, France',
 'Fédération Internationale de l'Automobile (FIA)',
 1950, 24),
('Formula 2',
 'FIA Formula 2 Championship',
 'Paris, France',

```

```

'Fédération Internationale de l'Automobile (FIA)',
2017, 14),
('Formula E',
'ABB FIA Formula E World Championship',
'Geneva, Switzerland',
'Fédération Internationale de l'Automobile (FIA)',
2014, 16);

-- F1 Teams (series_id=1) - constructors points = sum of both drivers
INSERT INTO racing_teams (name, principal_name, base_location, power_unit, points,
↪ series_id) VALUES
('McLaren F1 Team',      'Andrea Stella',      'Woking, UK',              'Mercedes',
↪ 291, 1),    -- 1st
('Ferrari',              'Fred Vasseur',      'Maranello, Italy',        'Ferrari',
↪ 249, 1),    -- 2nd
('Red Bull Racing',      'Christian Horner',   'Milton Keynes, UK',        'Ford RBPT',
↪ 206, 1),    -- 3rd
('Mercedes-AMG Petronas', 'Toto Wolff',         'Brackley, UK',            'Mercedes',
↪ 138, 1),    -- 4th
('Aston Martin Aramco',  'Mike Krack',         'Silverstone, UK',         'Honda',
↪ 96, 1),     -- 5th
('Haas F1 Team',          'Ayao Komatsu',       'Kannapolis, USA',         'Ferrari',
↪ 58, 1),     -- 6th
('Alpine F1 Team',        'Oliver Oakes',       'Enstone, UK',             'Renault',
↪ 49, 1),     -- 7th
('Williams Racing',       'James Vowles',       'Grove, UK',               'Mercedes',
↪ 17, 1),     -- 8th
('Racing Bulls',         'Laurent Mekies',     'Faenza, Italy',           'Ford RBPT',
↪ 14, 1),     -- 9th
('Cadillac Racing',       'Michael Andretti',   'Fishers, USA',            'Cadillac',
↪ 6, 1),      -- 10th
('Audi F1 Team',          'Mattia Binotto',     'Hinwil, Switzerland',     'Audi',
↪ 4, 1);      -- 11th

-- F2 Teams (series_id=2)
INSERT INTO racing_teams (name, principal_name, base_location, points, series_id)
↪ VALUES

```

```

('Prema Racing',          'René Rosin',          'Grisignano, Italy',
↪ 89, 2),
('Hitech Grand Prix',    'Oliver Oakes',    'Silverstone, UK',
↪ 74, 2),
('ART Grand Prix',       'Sébastien Philippe', 'Villeneuve-la-Garenne, France',
↪ 58, 2),
('DAMS',                 'François Sicard',   'Le Mans, France',
↪ 41, 2);

-- FE Teams (series_id=3)
INSERT INTO racing_teams (name, principal_name, base_location, points, series_id)
↪ VALUES
('Jaguar TCS Racing',    'James Barclay',    'Silverstone, UK',    201,
↪ 3),
('Porsche TAG Heuer Formula E', 'Florian Modlinger', 'Weissach, Germany',    178,
↪ 3),
('DS Penske',           'Mark Preston',     'Versailles, France',
↪ 134, 3),
('Nissan Formula E Team', 'Tommaso Volpe',    'Yokohama, Japan',
↪ 98, 3);

-- F1 Drivers (scoring: 25-18-15-12-10-8-6-4-2-1 per race, ~8 races done)
-- Team IDs: McLaren=1, Ferrari=2, Red Bull=3, Mercedes=4, Aston Martin=5, Haas=6,
↪ Alpine=7, Williams=8, Racing Bulls=9, Cadillac=10, Audi=11
INSERT INTO racing_drivers (first_name, last_name, racing_number, points, team_id)
↪ VALUES
('Oscar',               'Piastrri',        81, 168, 1),    -- P1 WDC
('Lando',               'Norris',          4, 123, 1),    -- P2
('Charles',             'Leclerc',         16, 141, 2),    -- P3
('Lewis',               'Hamilton',        44, 108, 2),    -- P4
('Max',                 'Verstappen',      1, 127, 3),    -- P5
('Isack',               'Hadjari',         6, 79, 3),      -- P6
('George',              'Russell',         63, 98, 4),    -- P7
('Kimi',                'Antonelli',       12, 40, 4),     -- P8
('Fernando',            'Alonso',          14, 62, 5),    -- P9
('Lance',               'Stroll',          18, 34, 5),    -- P10
('Esteban',             'Ocon',            31, 33, 6),    -- P11

```

```

('Oliver',      'Bearman',      87,  25, 6),    -- P12
('Pierre',      'Gasly',        10,  28, 7),    -- P13
('Franco',      'Colapinto',    43,  21, 7),    -- P14
('Alexander',    'Albon',        23,  10, 8),    -- P15
('Carlos',      'Sainz',        55,   7, 8),    -- P16
('Liam',         'Lawson',       30,   9, 9),    -- P17
('Arvid',        'Lindblad',     6,   5, 9),    -- P18
('Sergio',       'Perez',        11,   6,10),    -- P19
('Valtteri',     'Bottas',       77,   0,10),    -- P20
('Nico',         'Hulkenberg',    27,   4,11),    -- P21
('Gabriel',      'Bortoleto',     5,   0,11);    -- P22

```

-- F2 Drivers (~6 rounds done, scoring 10-9-8-7-6-5-4-3-2-1)

```

INSERT INTO racing_drivers (first_name, last_name, racing_number, points, team_id)
↪ VALUES
('Paul',        'Aron',          7,  56, 12),
('Ugo',         'Ugochukwu',     3,  33, 12),
('Jak',         'Crawford',      8,  44, 13),
('Isack',       'Hadjar',        6,  30, 13),
('James',       'Wharton',       1,  35, 14),
('Tuukka',      'Taponen',     2,  23, 14),
('Oliver',      'Bearman',       5,  28, 15),
('Victor',      'Martins',     6,  13, 15);

```

-- FE Drivers (~10 rounds done, scoring 25-18-15-12-10-8-6-4-2-1)

```

INSERT INTO racing_drivers (first_name, last_name, racing_number, points, team_id)
↪ VALUES
('Nick',        'Cassidy',     37, 118, 16),
('Mitch',       'Evans',       9,   83, 16),
('Pascal',      'Wehrlein',   94, 111, 17),
('Antonio Felix','da Costa',   13,  67, 17),
('Jean-Eric',   'Vergne',     25,  89, 18),
('Stoffel',     'Vandoorne',  4,   45, 18),
('Oliver',      'Rowland',    22,  62, 19),
('Norman',      'Nato',       17,  36, 19);

```


Приложение A8. RacingTeamRepository.java

```
package com.vyatsu.racing.repositories;

import com.vyatsu.racing.models.RacingTeam;
import org.springframework.data.repository.CrudRepository;
import org.springframework.stereotype.Repository;

@Repository
public interface RacingTeamRepository extends CrudRepository<RacingTeam, Long> {
}
```

Приложение A9. RacingSeriesRepository.java

```
package com.vyatsu.racing.repositories;

import com.vyatsu.racing.models.RacingSeries;
import org.springframework.data.repository.CrudRepository;
import org.springframework.stereotype.Repository;

@Repository
public interface RacingSeriesRepository extends CrudRepository<RacingSeries, Long> {
}
```

Приложение Б1. Веб-интерфейс: Стили (static/style.css)

```
:root {
  --base:      #1e1e2e; --mantle:  #181825; --crust:    #11111b;
  --surface0:  #313244; --surface1:#45475a; --overlay0:#6c7086; --overlay1:#7f849c;
  --text:      #cdd6f4; --subtext0:#a6adc8; --subtext1:#bac2de;
  --lavender:  #b4befe; --blue:     #89b4fa; --sapphire:#74c7ec; --sky:       #89dceb;
  --teal:      #94e2d5; --green:    #a6e3a1; --yellow:   #f9e2af; --peach:    #fab387;
  --mauve:     #cba6f7; --red:       #f38ba8; --maroon:   #eba0ac; --pink:     #f5c2e7;
  --rosewater: #f5e0dc;
  /* Status colors */
  --live:      #a6e3a1; --active:   #89b4fa; --saved:     #f9e2af; --error:     #f38ba8;
```

```

}

* { box-sizing: border-box; margin: 0; padding: 0; }

body {
  font-family: 'Segoe UI', system-ui, sans-serif;
  background: var(--base);
  color: var(--text);
  min-height: 100vh;
  display: flex;
  flex-direction: column;
}

/* - TOP BAR (telemetry-style header) - */
.topbar {
  background: var(--crust);
  border-bottom: 1px solid var(--surface0);
  height: 48px;
  display: flex;
  align-items: center;
  padding: 0 24px;
  gap: 24px;
  flex-shrink: 0;
}

.topbar-brand {
  color: var(--mauve);
  font-weight: 900;
  font-size: 13px;
  letter-spacing: 1.5px;
  text-transform: uppercase;
  text-decoration: none;
}

.topbar-nav {
  display: flex;
  gap: 4px;
}

.topbar-nav a {

```

```

    color: var(--overlay0);
    text-decoration: none;
    font-size: 11px;
    font-weight: 700;
    letter-spacing: 0.8px;
    text-transform: uppercase;
    padding: 4px 12px;
    border-left: 2px solid transparent;
    transition: color .15s, border-color .15s;
}

.topbar-nav a:hover { color: var(--text); }
.topbar-nav a.active { color: var(--blue); border-left-color: var(--blue); }
.topbar-status {
    margin-left: auto;
    display: flex;
    align-items: center;
    gap: 8px;
}

.status-dot {
    width: 7px;
    height: 7px;
    border-radius: 0;
    flex-shrink: 0;
}

.status-dot.live { background: var(--live); }
.status-dot.ready { background: var(--active); }
.status-label {
    font-size: 10px;
    font-weight: 700;
    letter-spacing: 1px;
    text-transform: uppercase;
    color: var(--overlay0);
}

/* - PAGE LAYOUT - */

.page-wrap {
    display: flex;

```

```

    flex: 1;
    overflow: hidden;
}

/* - LEFT SIDEBAR - */
.sidebar {
    width: 220px;
    background: var(--mantle);
    border-right: 1px solid var(--surface0);
    padding: 16px;
    flex-shrink: 0;
    overflow-y: auto;
}

.sidebar-section {
    margin-bottom: 16px;
}

.sidebar-label {
    font-size: 9px;
    font-weight: 700;
    letter-spacing: 1.2px;
    text-transform: uppercase;
    color: var(--overlay0);
    margin-bottom: 8px;
    padding-bottom: 4px;
    border-bottom: 1px solid var(--surface0);
}

.sidebar-btn {
    display: block;
    width: 100%;
    padding: 8px 12px;
    background: var(--surface0);
    border: none;
    color: var(--text);
    font-size: 11px;
    font-weight: 700;
    letter-spacing: 0.5px;
    text-transform: uppercase;

```

```

    text-decoration: none;
    cursor: pointer;
    transition: background .15s;
    margin-bottom: 4px;
}

.sidebar-btn:hover { background: var(--surface1); }
.sidebar-btn.primary { background: var(--blue); color: var(--crust); }
.sidebar-btn.primary:hover { background: var(--sapphire); }
.sidebar-btn.add { background: var(--mauve); color: var(--crust); }
.sidebar-btn.add:hover { background: var(--lavender); }
.sidebar-btn.danger { background: var(--red); color: var(--crust); }

/* - MAIN CONTENT - */
.main-content {
    flex: 1;
    padding: 20px 24px;
    overflow-y: auto;
    min-width: 0;
}

/* - RIGHT SIDEBAR - */
.sidebar-right {
    width: 240px;
    background: var(--mantle);
    border-left: 1px solid var(--surface0);
    padding: 16px;
    flex-shrink: 0;
    overflow-y: auto;
}

/* - TELEMETRY PANEL - */
.tlm-panel {
    background: var(--mantle);
    border-left: 3px solid var(--blue);
    padding: 12px 14px;
    margin-bottom: 12px;
}

```

```

.tlm-panel.f1 { border-left-color: var(--red); }
.tlm-panel.f2 { border-left-color: var(--peach); }
.tlm-panel.fe { border-left-color: var(--green); }
.tlm-panel.wec { border-left-color: var(--sapphire); }
.tlm-panel.indy { border-left-color: var(--yellow); }
.tlm-panel.moto { border-left-color: var(--mauve); }
.tlm-panel-header {
  font-size: 9px;
  font-weight: 700;
  letter-spacing: 1.2px;
  text-transform: uppercase;
  color: var(--overlay0);
  margin-bottom: 8px;
}
.tlm-panel-value {
  font-size: 22px;
  font-weight: 900;
  color: var(--text);
}
.tlm-panel-sub {
  font-size: 11px;
  color: var(--overlay1);
  margin-top: 2px;
}
.tlm-panel-value.mauve { color: var(--mauve); }
.tlm-panel-value.blue { color: var(--blue); }
.tlm-panel-value.teal { color: var(--teal); }
.tlm-panel-value.green { color: var(--green); }
.tlm-panel-value.peach { color: var(--peach); }
.tlm-panel-value.yellow { color: var(--yellow); }

/* - PAGE HEADER - */
.page-header {
  display: flex;
  align-items: center;
  justify-content: space-between;
  margin-bottom: 16px;
}

```

```

}
.underline.page-header h1 {
  font-size: 16px;
  font-weight: 900;
  letter-spacing: 0.5px;
  text-transform: uppercase;
  color: var(--text);
}
.underline.page-header .underline.subtitle {
  font-size: 10px;
  color: var(--overlay0);
  letter-spacing: 0.5px;
  margin-top: 2px;
}

/* - STATS GRID - */
.underline.stats-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(150px, 1fr));
  gap: 10px;
  margin-bottom: 20px;
}
.underline.stat-card {
  background: var(--mantle);
  border-left: 3px solid var(--surface1);
  padding: 12px 14px;
}
.underline.stat-card.underline.wide { grid-column: span 2; }
.underline.stat-card.underline.highlight { border-left-color: var(--blue); }
.underline.stat-label {
  font-size: 9px;
  font-weight: 700;
  letter-spacing: 1.2px;
  text-transform: uppercase;
  color: var(--overlay0);
  margin-bottom: 4px;
}

```

```

.stat-value {
  font-size: 22px;
  font-weight: 900;
}

.stat-value.mauve { color: var(--mauve); }
.stat-value.blue { color: var(--blue); }
.stat-value.teal { color: var(--teal); }
.stat-value.yellow { color: var(--yellow); }
.stat-value.peach { color: var(--peach); }
.stat-value.green { color: var(--green); }

.stat-sub {
  font-size: 10px;
  color: var(--overlay1);
  margin-top: 2px;
}

/* - DATA TABLE - */

.data-table {
  width: 100%;
  border-collapse: collapse;
  background: var(--mantle);
  margin-bottom: 20px;
}

.data-table th {
  background: var(--crust);
  color: var(--blue);
  font-size: 9px;
  font-weight: 700;
  letter-spacing: 1px;
  text-transform: uppercase;
  padding: 8px 10px;
  text-align: left;
  border-bottom: 1px solid var(--surface0);
}

.data-table td {
  padding: 7px 10px;
  font-size: 12px;
}

```



```

    border-bottom: 1px solid var(--surface0);
}
.data-table tr:last-child td { border-bottom: none; }
.data-table tr:hover td { background: rgba(255,255,255,.03); }
.data-table tr.lead td { background: rgba(249,226,175,.06); }

/* Series separator row */
.series-row td {
    background: var(--crust);
    padding: 6px 10px;
    font-weight: 800;
    font-size: 10px;
    letter-spacing: 1px;
    text-transform: uppercase;
    color: var(--text);
    border-left: 3px solid transparent;
}
.series-row.f1 td { border-left-color: var(--red); color: var(--red); }
.series-row.f2 td { border-left-color: var(--peach); color: var(--peach); }
.series-row.fe td { border-left-color: var(--green); color: var(--green); }
.series-row.wec td { border-left-color: var(--sapphire); color: var(--sapphire); }
.series-row.indy td { border-left-color: var(--yellow); color: var(--yellow); }
.series-row.moto td { border-left-color: var(--mauve); color: var(--mauve); }

/* - POSITION COLORS - */
.gold { color: var(--yellow); font-weight: 900; }
.silver { color: var(--subtext1); font-weight: 700; }
.bronze { color: var(--peach); font-weight: 700; }
.dim { color: var(--overlay0); }

/* - BADGES - */
.badge {
    padding: 2px 6px;
    font-size: 10px;
    font-weight: 700;
}
.badge.pu { background: var(--surface0); color: var(--mauve); }

```

```

.badge.num { background: var(--surface0); color: var(--text); font-weight: 900; }
.badge.series-badge { background: var(--surface0); color: var(--sky); }
.badge.green { background: var(--green); color: var(--crust); }

/* - TEXT HELPERS - */
.name { font-weight: 700; color: var(--lavender); }
.yellow { color: var(--yellow); font-weight: bold; }

/* - SECTION TITLE - */
.section-title {
  font-size: 11px;
  font-weight: 700;
  letter-spacing: 1px;
  text-transform: uppercase;
  color: var(--mauve);
  border-bottom: 1px solid var(--surface0);
  padding-bottom: 5px;
  margin: 24px 0 12px;
}

/* - POINTS GRID - */
.pts-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(70px, 1fr));
  gap: 6px;
  margin-bottom: 16px;
}
.pts-item {
  background: var(--mantle);
  padding: 6px 8px;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 2px;
}
.pts-item .pos {
  font-size: 9px;

```

```

    color: var(--overlay0);
    font-weight: 700;
    letter-spacing: 0.5px;
}

.pts-item .pts {
    font-size: 15px;
    font-weight: 900;
    color: var(--text);
}

.pts-item .pts.yellow { color: var(--yellow); }
.pts-item .pts.peach { color: var(--peach); }
.pts-item .pts.dim { color: var(--overlay0); }

/* - SERIES HEADER - */
.series-header {
    display: flex;
    justify-content: space-between;
    align-items: flex-end;
    border-bottom: 2px solid var(--surface0);
    padding-bottom: 8px;
    margin: 28px 0 12px;
}

.series-header.fl { border-bottom-color: var(--red); }
.series-header.f2 { border-bottom-color: var(--peach); }
.series-header.fe { border-bottom-color: var(--green); }
.series-header.wec { border-bottom-color: var(--sapphire); }
.series-header.indy { border-bottom-color: var(--yellow); }
.series-header.moto { border-bottom-color: var(--mauve); }

.series-name {
    font-size: 14px;
    font-weight: 900;
    letter-spacing: 0.5px;
    text-transform: uppercase;
    color: var(--text);
}

.series-header.fl .series-name { color: var(--red); }
.series-header.f2 .series-name { color: var(--peach); }

```

```

.series-header.fe .series-name { color: var(--green); }
.series-official {
  font-size: 10px;
  color: var(--overlay0);
  margin-top: 2px;
}
.series-pills { display: flex; gap: 6px; align-items: center; }
.pill {
  background: var(--surface0);
  color: var(--subtext1);
  font-size: 9px;
  padding: 3px 8px;
  font-weight: 700;
  letter-spacing: 0.3px;
}

.team-accent {
  width: 3px;
  height: 14px;
  display: inline-block;
  vertical-align: middle;
  margin-right: 6px;
}

/* - BUTTONS - */
.btn {
  display: inline-block;
  padding: 6px 14px;
  font-size: 10px;
  font-weight: 700;
  letter-spacing: 0.5px;
  text-transform: uppercase;
  border: none;
  cursor: pointer;
  text-decoration: none;
  transition: opacity .15s;
  color: var(--crust);
}

```

```

    background: var(--blue);
}
.btn:hover { opacity: .8; }
.btn-add { background: var(--green); }
.btn-save { background: var(--blue); }
.btn-ghost { background: var(--surface1); color: var(--text); }
.btn-sm { padding: 3px 8px; font-size: 9px; }
.btn.edit { background: var(--yellow); color: var(--crust); }
.btn.del { background: var(--red); }
.btn-group { display: flex; gap: 4px; }

/* - FILTER BAR - */
.filter-bar {
    display: flex;
    gap: 8px;
    align-items: center;
    margin-bottom: 14px;
    flex-wrap: wrap;
    background: var(--mantle);
    padding: 10px 14px;
    border-left: 3px solid var(--surface1);
}
.filter-select, .filter-input {
    background: var(--surface0);
    border: 1px solid var(--surface1);
    color: var(--text);
    padding: 6px 10px;
    font-size: 11px;
}

/* - FORM - */
.form-card {
    background: var(--mantle);
    border-left: 3px solid var(--blue);
    padding: 24px;
    max-width: 640px;
}

```

```

.form-row { display: grid; grid-template-columns: 1fr 1fr; gap: 14px; }
.form-group { margin-bottom: 16px; }
.form-group label {
  display: block;
  font-size: 9px;
  font-weight: 700;
  letter-spacing: 1px;
  text-transform: uppercase;
  color: var(--overlay0);
  margin-bottom: 4px;
}
.form-group input, .form-group select, .form-group textarea {
  width: 100%;
  background: var(--surface0);
  border: 1px solid var(--surface1);
  color: var(--text);
  padding: 7px 10px;
  font-size: 12px;
}
.form-group input:focus, .form-group select:focus {
  outline: none;
  border-color: var(--blue);
}
.form-actions { display: flex; gap: 8px; margin-top: 12px; }

/* - ERROR BOX - */
.error-box {
  background: rgba(243,139,168,.1);
  border-left: 3px solid var(--red);
  padding: 10px 14px;
  margin-bottom: 16px;
  color: var(--red);
}
.error-box p { margin: 2px 0; font-size: 11px; }

/* - ACTIONS COLUMN - */
.actions { white-space: nowrap; }

```

```
.empty { text-align: center; color: var(--overlay0); padding: 20px; font-size: 12px;
↪ }
```

```
/* - SERIES OVERVIEW GRID - */
```

```
.series-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(200px, 1fr));
  gap: 10px;
  margin-bottom: 20px;
}
.series-card {
  background: var(--mantle);
  border-left: 3px solid var(--blue);
  padding: 12px 14px;
}
.series-card.f1 { border-left-color: var(--red); }
.series-card.f2 { border-left-color: var(--peach); }
.series-card.fe { border-left-color: var(--green); }
.series-card.wec { border-left-color: var(--sapphire); }
.series-card.indy { border-left-color: var(--yellow); }
.series-card.moto { border-left-color: var(--mauve); }
.sc-name { font-size: 13px; font-weight: 800; letter-spacing: 0.3px; }
.sc-official { font-size: 9px; color: var(--overlay0); margin: 3px 0 8px; }
.sc-meta { display: flex; flex-wrap: wrap; gap: 6px; }
.sc-meta span {
  background: var(--surface0);
  padding: 2px 8px;
  font-size: 9px;
  color: var(--subtext0);
}
```

```
/* - BOTTOM STATUS BAR - */
```

```
.bottombar {
  background: var(--crust);
  border-top: 1px solid var(--surface0);
  height: 32px;
  display: flex;
```

```

    align-items: center;
    padding: 0 24px;
    gap: 16px;
    flex-shrink: 0;
    font-size: 9px;
    color: var(--overlay0);
    letter-spacing: 0.5px;
}
.bottombar .stat {
    display: flex;
    align-items: center;
    gap: 5px;
}
.bottombar .stat-dot {
    width: 5px;
    height: 5px;
    background: var(--green);
}
.bottombar .stat-label-bot {
    color: var(--overlay1);
    text-transform: uppercase;
    font-weight: 700;
}
.bottombar .stat-value-bot {
    color: var(--subtext0);
}
.bottombar .db {
    margin-left: auto;
}

```

/ - UTILITY - */*

```

.flex { display: flex; }
.gap-2 { gap: 8px; }
.gap-4 { gap: 16px; }
.items-center { align-items: center; }
.justify-between { justify-content: space-between; }
.mt-2 { margin-top: 8px; }

```



```

.mb-2 { margin-bottom: 8px; }
.text-sm { font-size: 10px; }
.text-overlay { color: var(--overlay0); }

/* - MOBILE - */
@media (max-width: 900px) {
  .sidebar { display: none; }
  .sidebar-right { display: none; }
  .page-wrap { flex-direction: column; }
  .main-content { padding: 16px; }
  .topbar { padding: 0 12px; gap: 12px; }
  .topbar-nav { display: none; }
  .stats-grid { grid-template-columns: 1fr 1fr; }
  .stat-card.wide { grid-column: span 2; }
  .series-grid { grid-template-columns: 1fr; }
  .form-row { grid-template-columns: 1fr; gap: 0; }
  .form-card { padding: 16px; }
}

```

Приложение Б2. Главная страница (templates/index.html)

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>DASHBOARD - Racing Management</title>
<link rel="stylesheet" th:href="@{/style.css}">
</head>
<body>

<div class="topbar">
  <a href="/" class="topbar-brand">RACING SYSTEM</a>
  <nav class="topbar-nav">
    <a href="/" class="active">DASHBOARD</a>
    <a href="/series">SERIES</a>
    <a href="/teams">TEAMS</a>

```

```

    <a href="/drivers">DRIVERS</a>
</nav>
<div class="topbar-status">
    <span class="status-dot live"></span>
    <span class="status-label">LIVE</span>
</div>
</div>

<div class="page-wrap">

    <aside class="sidebar">
        <div class="sidebar-section">
            <div class="sidebar-label">Navigation</div>
            <a href="/" class="sidebar-btn primary">DASHBOARD</a>
            <a href="/series" class="sidebar-btn">SERIES</a>
            <a href="/teams" class="sidebar-btn">TEAMS</a>
            <a href="/drivers" class="sidebar-btn">DRIVERS</a>
        </div>
        <div class="sidebar-section">
            <div class="sidebar-label">Quick Actions</div>
            <a href="/series/add" class="sidebar-btn add">+ ADD SERIES</a>
            <a href="/teams/add" class="sidebar-btn add">+ ADD TEAM</a>
            <a href="/drivers/add" class="sidebar-btn add">+ ADD DRIVER</a>
        </div>
    </aside>

    <main class="main-content">

        <div class="page-header">
            <div>
                <h1>RACING WORLD MANAGEMENT</h1>
                <p class="subtitle">F1 · F2 · FE · WEC · INDYCAR · MotoGP - 2026 SEASON</p>
            </div>
        </div>

        <div class="stats-grid">
            <div class="stat-card highlight">

```

```

    <div class="stat-label">Series</div>
    <div class="stat-value mauve" th:text="${seriesList.size()}"></div>
</div>
<div class="stat-card highlight">
    <div class="stat-label">Teams</div>
    <div class="stat-value blue" th:text="${teams.size()}"></div>
</div>
<div class="stat-card highlight">
    <div class="stat-label">Drivers</div>
    <div class="stat-value teal" th:text="${drivers.size()}"></div>
</div>
<div class="stat-card wide highlight">
    <div class="stat-label">Championship Leader</div>
    <div class="stat-value peach" th:text="${drivers[0].firstName + ' ' +
↳ drivers[0].lastName}"></div>
    <div class="stat-sub" th:text="${drivers[0].points + ' pts - ' +
↳ drivers[0].team.name}"></div>
</div>
</div>

<div class="section-title">F1 Points System</div>
<div class="pts-grid">
    <div class="pts-item"><span class="pos">P1</span><span class="pts
↳ yellow">25</span></div>
    <div class="pts-item"><span class="pos">P2</span><span
↳ class="pts">18</span></div>
    <div class="pts-item"><span class="pos">P3</span><span class="pts
↳ peach">15</span></div>
    <div class="pts-item"><span class="pos">P4</span><span
↳ class="pts">12</span></div>
    <div class="pts-item"><span class="pos">P5</span><span
↳ class="pts">10</span></div>
    <div class="pts-item"><span class="pos">P6</span><span
↳ class="pts">8</span></div>
    <div class="pts-item"><span class="pos">P7</span><span
↳ class="pts">6</span></div>

```

```

<div class="pts-item"><span class="pos">P8</span><span
↳ class="pts">4</span></div>
<div class="pts-item"><span class="pos">P9</span><span
↳ class="pts">2</span></div>
<div class="pts-item"><span class="pos">P10</span><span
↳ class="pts">1</span></div>
</div>

<div class="section-title">Series Overview</div>
<div class="series-grid">
  <th:block th:each="series : ${seriesList}">
    <div class="series-card"
      th:classappend="${series.name == 'Formula 1' ? 'f1' : (series.name ==
↳ 'Formula 2' ? 'f2' : (series.name == 'Formula E' ? 'fe' :
↳ (series.name == 'WEC' ? 'wec' : (series.name == 'IndyCar' ? 'indy' :
↳ (series.name == 'MotoGP' ? 'moto' : ''))))))}">
      <div class="sc-name" th:text="${series.name}"></div>
      <div class="sc-official" th:text="${series.officialName}"></div>
      <div class="sc-meta">
        <span th:text="'Founded ' + ${series.foundedYear}"></span>
        <span th:text="${series.rounds} + ' rounds'"></span>
        <span th:text="${series.headquarters}"></span>
      </div>
    </div>
  </th:block>
</div>

<div class="section-title" style="color:var(--peach)">Drivers Championship
↳ Standings</div>
<table class="data-table">
  <thead>
    <tr><th>Pos</th><th>Driver</th><th>No.</th><th>Team</th><th>Points</th></tr>
  </thead>
  <tbody>
    <th:block th:each="series : ${seriesList}">
      <tr class="series-row"

```

```

        th:classappend="${series.name == 'Formula 1' ? 'f1' : (series.name ==
        ↪ 'Formula 2' ? 'f2' : (series.name == 'Formula E' ? 'fe' :
        ↪ (series.name == 'WEC' ? 'wec' : (series.name == 'IndyCar' ? 'indy'
        ↪ : (series.name == 'MotoGP' ? 'moto' : ''))))))}">
        <td colspan="5" th:text="${series.name + ' CHAMPIONSHIP'}"></td>
    </tr>
    <th:block th:each="entry : ${standingsBySeries.entrySet()}">
        <th:block th:if="${entry.key.id == series.id}">
            <tr th:each="driver, iter : ${entry.value}"
            ↪ th:classappend="${iter.index == 0} ? 'leader' : ''">
                <td th:classappend="${iter.index == 0} ? 'gold' : (${iter.index == 1}
                ↪ ? 'silver' : (${iter.index == 2} ? 'bronze' : 'dim'))"
                    th:text="${iter.count}"></td>
                <td class="name">
                    <span class="team-accent" th:style="'background-color:' +
                    ↪ ${driver.team.teamColor}"></span>
                    <span th:text="${driver.firstName + ' ' + driver.lastName}"></span>
                </td>
                <td><span class="badge num" th:text="'#' +
                ↪ ${driver.racingNumber}"></span></td>
                <td th:text="${driver.team.name}"></td>
                <td class="pts yellow" th:text="${driver.points}"></td>
            </tr>
        </th:block>
    </th:block>
</tbody>
</table>

</main>

<aside class="sidebar-right">
    <div class="tlm-panel">
        <div class="tlm-panel-header">System Status</div>
        <div class="tlm-panel-value green" style="font-size:14px">ONLINE</div>
        <div class="tlm-panel-sub">PostgreSQL connected</div>
    </div>

```

```

<div class="tlm-panel">
  <div class="tlm-panel-header">Total Points Awarded</div>
  <div class="tlm-panel-value yellow" style="font-size:20px"
    ↪ th:text="${drivers.size()}"></div>
  <div class="tlm-panel-sub">registered drivers</div>
</div>
<div class="tlm-panel">
  <div class="tlm-panel-header">Active Series</div>
  <div class="tlm-panel-value mauve" style="font-size:20px"
    ↪ th:text="${seriesList.size()}"></div>
  <div class="tlm-panel-sub">racing championships</div>
</div>
</aside>

</div>

<div class="bottombar">
  <div class="stat">
    <span class="stat-dot"></span>
    <span class="stat-label-bot">DB</span>
    <span class="stat-value-bot">LIVE</span>
  </div>
  <div class="stat">
    <span class="stat-label-bot">Series</span>
    <span class="stat-value-bot" th:text="${seriesList.size()}"></span>
  </div>
  <div class="stat">
    <span class="stat-label-bot">Teams</span>
    <span class="stat-value-bot" th:text="${teams.size()}"></span>
  </div>
  <div class="stat">
    <span class="stat-label-bot">Drivers</span>
    <span class="stat-value-bot" th:text="${drivers.size()}"></span>
  </div>
  <div class="db">Java / Spring Boot 2.6 / Thymeleaf</div>
</div>

```

```
</body>
</html>
```

Приложение Б3. Чемпионаты (templates/series.html)

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SERIES - Racing Management</title>
<link rel="stylesheet" th:href="@{/style.css}">
</head>
<body>

<div class="topbar">
  <a href="/" class="topbar-brand">RACING SYSTEM</a>
  <nav class="topbar-nav">
    <a href="/">DASHBOARD</a>
    <a href="/series" class="active">SERIES</a>
    <a href="/teams">TEAMS</a>
    <a href="/drivers">DRIVERS</a>
  </nav>
  <div class="topbar-status">
    <span class="status-dot live"></span>
    <span class="status-label">LIVE</span>
  </div>
</div>

<div class="page-wrap">

  <aside class="sidebar">
    <div class="sidebar-section">
      <div class="sidebar-label">Navigation</div>
      <a href="/" class="sidebar-btn">DASHBOARD</a>
      <a href="/series" class="sidebar-btn primary">SERIES</a>
      <a href="/teams" class="sidebar-btn">TEAMS</a>
```

```

    <a href="/drivers" class="sidebar-btn">DRIVERS</a>
</div>
<div class="sidebar-section">
    <div class="sidebar-label">Actions</div>
    <a href="/series/add" class="sidebar-btn add">+ ADD SERIES</a>
</div>
</aside>

<main class="main-content">

<div class="page-header">
    <h1>RACING SERIES</h1>
    <a href="/series/add" class="btn btn-add">+ ADD SERIES</a>
</div>

<table class="data-table">
    <thead>
        <tr>
            <th>Name</th><th>Official Name</th><th>HQ</th><th>Governing
                ↳ Body</th><th>Founded</th><th>Rounds</th><th>Teams</th><th>Actions</th>
        </tr>
    </thead>
    <tbody>
        <tr th:each="series : ${seriesList}">
            <td class="name"><span th:text="${series.name}"></span></td>
            <td th:text="${series.officialName}"></td>
            <td th:text="${series.headquarters}"></td>
            <td th:text="${series.governingBody}"></td>
            <td th:text="${series.foundedYear}"></td>
            <td th:text="${series.rounds}"></td>
            <td th:text="${series.teams.size()}"></td>
            <td class="actions">
                <a th:href="@{/series/edit/{id}(id=${series.id})}" class="btn btn-sm
                    ↳ edit">Edit</a>
                <a th:href="@{/series/delete/{id}(id=${series.id})}" class="btn btn-sm
                    ↳ del"

```



```

        onclick="return confirm('Delete this series and all its
        ↪ teams/drivers?');">Delete</a>
    </td>
</tr>
</tbody>
</table>

</main>

<aside class="sidebar-right">
    <div class="tlm-panel">
        <div class="tlm-panel-header">Total Series</div>
        <div class="tlm-panel-value mauve" style="font-size:20px"
        ↪ th:text="${seriesList.size()}"></div>
        <div class="tlm-panel-sub">racing championships managed</div>
    </div>
</aside>

</div>

<div class="bottombar">
    <div class="stat">
        <span class="stat-dot"></span>
        <span class="stat-label-bot">DB</span>
        <span class="stat-value-bot">LIVE</span>
    </div>
    <div class="stat">
        <span class="stat-label-bot">Series</span>
        <span class="stat-value-bot" th:text="${seriesList.size()}"></span>
    </div>
    <div class="db">Java / Spring Boot 2.6 / Thymeleaf</div>
</div>

</body>
</html>

```

Приложение Б4. Команды (templates/teams.html)

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>TEAMS - Racing Management</title>
<link rel="stylesheet" th:href="@{/style.css}">
</head>
<body>

<div class="topbar">
  <a href="/" class="topbar-brand">RACING SYSTEM</a>
  <nav class="topbar-nav">
    <a href="/">DASHBOARD</a>
    <a href="/series">SERIES</a>
    <a href="/teams" class="active">TEAMS</a>
    <a href="/drivers">DRIVERS</a>
  </nav>
  <div class="topbar-status">
    <span class="status-dot live"></span>
    <span class="status-label">LIVE</span>
  </div>
</div>

<div class="page-wrap">

  <aside class="sidebar">
    <div class="sidebar-section">
      <div class="sidebar-label">Navigation</div>
      <a href="/" class="sidebar-btn">DASHBOARD</a>
      <a href="/series" class="sidebar-btn">SERIES</a>
      <a href="/teams" class="sidebar-btn primary">TEAMS</a>
      <a href="/drivers" class="sidebar-btn">DRIVERS</a>
    </div>
    <div class="sidebar-section">
```

```

    <div class="sidebar-label">Actions</div>
    <a href="/teams/add" class="sidebar-btn add">+ ADD TEAM</a>
  </div>
</aside>

<main class="main-content">

  <div class="page-header">
    <h1>RACING TEAMS</h1>
    <a href="/teams/add" class="btn btn-add">+ ADD TEAM</a>
  </div>

  <div th:each="series : ${seriesList}">
    <div class="series-header"
      th:classappend="${series.name == 'Formula 1' ? 'f1' : (series.name ==
        ↪ 'Formula 2' ? 'f2' : (series.name == 'Formula E' ? 'fe' : (series.name
        ↪ == 'WEC' ? 'wec' : (series.name == 'IndyCar' ? 'indy' : (series.name
        ↪ == 'MotoGP' ? 'moto' : ''))))})">
      <div>
        <div class="series-name" th:text="${series.name}"></div>
        <div class="series-official" th:text="${series.officialName}"></div>
      </div>
      <div class="series-pills">
        <div class="pill">Founded <b th:text="${series.foundedYear}"></b></div>
        <div class="pill"><b th:text="${series.rounds}"></b> Rounds</div>
        <div class="pill" th:text="${series.headquarters}"></div>
      </div>
    </div>

    <table class="data-table">
      <thead>
        <tr>
          <th>Team</th><th>Principal</th><th>HQ</th><th>Power
            ↪ Unit</th><th>Points</th><th>Wins</th><th>Podiums</th><th>Actions</th>
        </tr>
      </thead>
      <tbody>

```

```

<tr th:each="team : ${series.teams}">
  <td class="name">
    <span class="team-accent" th:style="'background-color:' +
      ↪ ${team.teamColor}"></span>
    <span th:text="${team.name}"></span>
  </td>
  <td th:text="${team.principalName}"></td>
  <td th:text="${team.baseLocation}"></td>
  <td><span class="badge pu" th:text="${team.powerUnit ? :
    ↪ '-'}"></span></td>
  <td class="pts yellow" th:text="${team.points}"></td>
  <td th:text="${team.wins}"></td>
  <td th:text="${team.podiums}"></td>
  <td class="actions">
    <a th:href="@{/teams/edit/{id}(id=${team.id})}" class="btn btn-sm
      ↪ edit">Edit</a>
    <a th:href="@{/teams/delete/{id}(id=${team.id})}" class="btn btn-sm
      ↪ del"
      onclick="return confirm('Delete this team?');">Delete</a>
  </td>
</tr>
</tbody>
</table>
</div>

</main>

<aside class="sidebar-right">
  <div class="tlm-panel">
    <div class="tlm-panel-header">Total Teams</div>
    <div class="tlm-panel-value mauve" style="font-size:20px">-</div>
    <div class="tlm-panel-sub">across all series</div>
  </div>
  <div class="tlm-panel">
    <div class="tlm-panel-header">Series</div>
    <div class="tlm-panel-value blue" style="font-size:20px"
      ↪ th:text="${seriesList.size()}"></div>

```

```

        <div class="tlm-panel-sub">racing championships</div>
    </div>
</aside>

</div>

<div class="bottombar">
    <div class="stat">
        <span class="stat-dot"></span>
        <span class="stat-label-bot">DB</span>
        <span class="stat-value-bot">LIVE</span>
    </div>
    <div class="stat">
        <span class="stat-label-bot">Series</span>
        <span class="stat-value-bot" th:text="${seriesList.size()}"></span>
    </div>
    <div class="db">Java / Spring Boot 2.6 / Thymeleaf</div>
</div>

</body>
</html>

```

Приложение Б5. Пилоты (templates/drivers.html)

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>DRIVERS - Racing Management</title>
<link rel="stylesheet" th:href="@{/style.css}">
</head>
<body>

<div class="topbar">
    <a href="/" class="topbar-brand">RACING SYSTEM</a>
    <nav class="topbar-nav">

```

```

    <a href="/">DASHBOARD</a>
    <a href="/series">SERIES</a>
    <a href="/teams">TEAMS</a>
    <a href="/drivers" class="active">DRIVERS</a>
</nav>
<div class="topbar-status">
    <span class="status-dot live"></span>
    <span class="status-label">LIVE</span>
</div>
</div>

<div class="page-wrap">

    <aside class="sidebar">
        <div class="sidebar-section">
            <div class="sidebar-label">Navigation</div>
            <a href="/" class="sidebar-btn">DASHBOARD</a>
            <a href="/series" class="sidebar-btn">SERIES</a>
            <a href="/teams" class="sidebar-btn">TEAMS</a>
            <a href="/drivers" class="sidebar-btn primary">DRIVERS</a>
        </div>
        <div class="sidebar-section">
            <div class="sidebar-label">Actions</div>
            <a href="/drivers/add" class="sidebar-btn add">+ ADD DRIVER</a>
        </div>
    </aside>

    <main class="main-content">

        <div class="page-header">
            <h1>ALL DRIVERS</h1>
            <a href="/drivers/add" class="btn btn-add">+ ADD DRIVER</a>
        </div>

        <table class="data-table">
            <thead>
                <tr>

```

```

        ↪ <th>#</th><th>Driver</th><th>No.</th><th>Pts</th><th>Team</th><th>Actions</th>
    </tr>
</thead>
<tbody>
    <th:block th:each="series : ${seriesList}">
        <tr class="series-row"
            th:classappend="${series.name == 'Formula 1' ? 'f1' : (series.name ==
                ↪ 'Formula 2' ? 'f2' : (series.name == 'Formula E' ? 'fe' :
                ↪ (series.name == 'WEC' ? 'wec' : (series.name == 'IndyCar' ? 'indy'
                ↪ : (series.name == 'MotoGP' ? 'moto' : ''))))))}">
            <td colspan="6" th:text="${series.name + ' - ' +
                ↪ series.officialName}"></td>
        </tr>
        <th:block th:each="team : ${series.teams}">
            <tr th:each="driver : ${team.drivers}">
                <td th:text="${driver.id}" class="dim"></td>
                <td class="name">
                    <span class="team-accent" th:style="'background-color:' +
                        ↪ ${team.teamColor}"></span>
                    <span th:text="${driver.firstName + ' ' + driver.lastName}"></span>
                </td>
                <td><span class="badge num" th:text="'#' +
                    ↪ ${driver.racingNumber}"></span></td>
                <td class="pts yellow" th:text="${driver.points}"></td>
                <td th:text="${team.name}"></td>
                <td class="actions">
                    <a th:href="@{/drivers/edit/{id}(id=${driver.id})}" class="btn btn-sm
                        ↪ edit">Edit</a>
                    <a th:href="@{/drivers/delete/{id}(id=${driver.id})}" class="btn
                        ↪ btn-sm del"
                        onclick="return confirm('Delete this driver?');">Delete</a>
                </td>
            </tr>
        </th:block>
    </th:block>
</tbody>

```

```

</table>

</main>

<aside class="sidebar-right">
  <div class="tlm-panel">
    <div class="tlm-panel-header">Series</div>
    <div class="tlm-panel-value blue" style="font-size:20px"
      ↪ th:text="${seriesList.size()}"></div>
    <div class="tlm-panel-sub">racing championships</div>
  </div>
</aside>

</div>

<div class="bottombar">
  <div class="stat">
    <span class="stat-dot"></span>
    <span class="stat-label-bot">DB</span>
    <span class="stat-value-bot">LIVE</span>
  </div>
  <div class="stat">
    <span class="stat-label-bot">Series</span>
    <span class="stat-value-bot" th:text="${seriesList.size()}"></span>
  </div>
  <div class="db">Java / Spring Boot 2.6 / Thymeleaf</div>
</div>

</body>
</html>

```

Приложение Б6. Форма чемпионата (templates/series-form.html)

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8">

```



```

<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title th:text="${series.id == null} ? 'NEW SERIES - Racing' : 'EDIT SERIES -
↳ Racing'"></title>
<link rel="stylesheet" th:href="@{/style.css}">
</head>
<body>

<div class="topbar">
  <a href="/" class="topbar-brand">RACING SYSTEM</a>
  <nav class="topbar-nav">
    <a href="/">DASHBOARD</a>
    <a href="/series" class="active">SERIES</a>
    <a href="/teams">TEAMS</a>
    <a href="/drivers">DRIVERS</a>
  </nav>
  <div class="topbar-status">
    <span class="status-dot live"></span>
    <span class="status-label">LIVE</span>
  </div>
</div>

<div class="page-wrap">

  <aside class="sidebar">
    <div class="sidebar-section">
      <div class="sidebar-label">Navigation</div>
      <a href="/" class="sidebar-btn">DASHBOARD</a>
      <a href="/series" class="sidebar-btn primary">SERIES</a>
      <a href="/teams" class="sidebar-btn">TEAMS</a>
      <a href="/drivers" class="sidebar-btn">DRIVERS</a>
    </div>
  </aside>

  <main class="main-content">

    <div class="page-header">
      <h1 th:text="${series.id == null} ? 'NEW SERIES' : 'EDIT SERIES'"></h1>

```

```

    <a href="/series" class="btn btn-ghost">← BACK</a>
</div>

<form class="form-card" action="#" th:action="@{/series/save}"
↪  th:object="${series}" method="post">
    <input type="hidden" th:field="*{id}" />

    <div class="form-row">
        <div class="form-group">
            <label>Name *</label>
            <input type="text" th:field="*{name}" required />
        </div>
        <div class="form-group">
            <label>Official Name</label>
            <input type="text" th:field="*{officialName}" />
        </div>
    </div>

    <div class="form-row">
        <div class="form-group">
            <label>Headquarters *</label>
            <input type="text" th:field="*{headquarters}" required />
        </div>
        <div class="form-group">
            <label>Governing Body</label>
            <input type="text" th:field="*{governingBody}" />
        </div>
    </div>

    <div class="form-row">
        <div class="form-group">
            <label>Founded Year</label>
            <input type="number" th:field="*{foundedYear}" />
        </div>
        <div class="form-group">
            <label>Rounds</label>
            <input type="number" th:field="*{rounds}" />
        </div>
    </div>

```

```

        </div>
    </div>

    <div class="form-actions">
        <button type="submit" class="btn btn-save">SAVE SERIES</button>
        <a href="/series" class="btn btn-ghost">CANCEL</a>
    </div>
</form>

</main>

</div>

<div class="bottombar">
    <div class="stat">
        <span class="stat-dot"></span>
        <span class="stat-label-bot">DB</span>
        <span class="stat-value-bot">LIVE</span>
    </div>
    <div class="stat">
        <span class="stat-label-bot">Mode</span>
        <span class="stat-value-bot">EDITOR</span>
    </div>
    <div class="db">Java / Spring Boot 2.6 / Thymeleaf</div>
</div>

</body>
</html>

```

Приложение Б7. Форма команды (templates/team-form.html)

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

```

```

<title th:text="${team.id == null} ? 'NEW TEAM - Racing' : 'EDIT TEAM -
↳ Racing'"></title>
<link rel="stylesheet" th:href="@{/style.css}">
</head>
<body>

<div class="topbar">
  <a href="/" class="topbar-brand">RACING SYSTEM</a>
  <nav class="topbar-nav">
    <a href="/">DASHBOARD</a>
    <a href="/series">SERIES</a>
    <a href="/teams" class="active">TEAMS</a>
    <a href="/drivers">DRIVERS</a>
  </nav>
  <div class="topbar-status">
    <span class="status-dot live"></span>
    <span class="status-label">LIVE</span>
  </div>
</div>

<div class="page-wrap">

  <aside class="sidebar">
    <div class="sidebar-section">
      <div class="sidebar-label">Navigation</div>
      <a href="/" class="sidebar-btn">DASHBOARD</a>
      <a href="/series" class="sidebar-btn">SERIES</a>
      <a href="/teams" class="sidebar-btn primary">TEAMS</a>
      <a href="/drivers" class="sidebar-btn">DRIVERS</a>
    </div>
  </aside>

  <main class="main-content">

    <div class="page-header">
      <h1 th:text="${team.id == null} ? 'NEW TEAM' : 'EDIT TEAM'"></h1>
      <a href="/teams" class="btn btn-ghost">← BACK</a>
    </div>
  </main>
</div>

```

```

</div>

<form class="form-card" action="#" th:action="@{/teams/save}"
↳ th:object="${team}" method="post">
  <input type="hidden" th:field="*{id}" />

  <div class="form-group">
    <label>Series *</label>
    <select th:field="*{series.id}" required>
      <option value="">- SELECT SERIES </option>
      <option th:each="ser : ${seriesList}"
        th:value="${ser.id}"
        th:text="${ser.name}"></option>
    </select>
  </div>

  <div class="form-group">
    <label>Team Name *</label>
    <input type="text" th:field="*{name}" required />
  </div>

  <div class="form-group">
    <label>Principal Name *</label>
    <input type="text" th:field="*{principalName}" required />
  </div>

  <div class="form-group">
    <label>Base Location *</label>
    <input type="text" th:field="*{baseLocation}" required />
  </div>

  <div class="form-group">
    <label>Power Unit</label>
    <input type="text" th:field="*{powerUnit}" />
  </div>

  <div class="form-group">

```

```

    <label>Points</label>
    <input type="number" th:field="*{points}" required />
</div>

<div class="form-row">
    <div class="form-group">
        <label>Wins</label>
        <input type="number" th:field="*{wins}" />
    </div>
    <div class="form-group">
        <label>Podiums</label>
        <input type="number" th:field="*{podiums}" />
    </div>
</div>

<div class="form-row">
    <div class="form-group">
        <label>Budget Cap (M)</label>
        <input type="number" step="0.1" th:field="*{budgetCap}" />
    </div>
    <div class="form-group">
        <label>Constructor Pos</label>
        <input type="number" th:field="*{constructorPos}" />
    </div>
</div>

<div class="form-row">
    <div class="form-group">
        <label>Chassis Model</label>
        <input type="text" th:field="*{chassisModel}" />
    </div>
    <div class="form-group">
        <label>Graduates</label>
        <input type="number" th:field="*{graduates}" />
    </div>
</div>

```

```

<div class="form-row">
  <div class="form-group">
    <label>Energy Partner</label>
    <input type="text" th:field="*{energyPartner}" />
  </div>
  <div class="form-group">
    <label>Battery kWh</label>
    <input type="number" step="0.1" th:field="*{batteryKwh}" />
  </div>
</div>

<div class="form-row">
  <div class="form-group">
    <label>Sustainability Score</label>
    <input type="number" th:field="*{sustainScore}" />
  </div>
  <div class="form-group">
    <label>Is Feeder</label>
    <select th:field="*{isFeeder}">
      <option th:value="false" th:selected="{team.isFeeder ==
        ↪ false}">No</option>
      <option th:value="true" th:selected="{team.isFeeder ==
        ↪ true}">Yes</option>
    </select>
  </div>
</div>

<div class="form-actions">
  <button type="submit" class="btn btn-save">SAVE TEAM</button>
  <a href="/teams" class="btn btn-ghost">CANCEL</a>
</div>
</form>

</main>

</div>

```

```

<div class="bottombar">
  <div class="stat">
    <span class="stat-dot"></span>
    <span class="stat-label-bot">DB</span>
    <span class="stat-value-bot">LIVE</span>
  </div>
  <div class="stat">
    <span class="stat-label-bot">Mode</span>
    <span class="stat-value-bot">EDITOR</span>
  </div>
  <div class="db">Java / Spring Boot 2.6 / Thymeleaf</div>
</div>

</body>
</html>

```

Приложение Б8. Форма пилота (templates/driver-form.html)

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title th:text="${driver.id == null} ? 'NEW DRIVER - Racing' : 'EDIT DRIVER -
↳ Racing'"></title>
<link rel="stylesheet" th:href="@{/style.css}">
</head>
<body>

<div class="topbar">
  <a href="/" class="topbar-brand">RACING SYSTEM</a>
  <nav class="topbar-nav">
    <a href="/">DASHBOARD</a>
    <a href="/series">SERIES</a>
    <a href="/teams">TEAMS</a>
    <a href="/drivers" class="active">DRIVERS</a>
  </nav>

```



```

<div class="topbar-status">
  <span class="status-dot live"></span>
  <span class="status-label">LIVE</span>
</div>
</div>

<div class="page-wrap">

  <aside class="sidebar">
    <div class="sidebar-section">
      <div class="sidebar-label">Navigation</div>
      <a href="/" class="sidebar-btn">DASHBOARD</a>
      <a href="/series" class="sidebar-btn">SERIES</a>
      <a href="/teams" class="sidebar-btn">TEAMS</a>
      <a href="/drivers" class="sidebar-btn primary">DRIVERS</a>
    </div>
  </aside>

  <main class="main-content">

    <div class="page-header">
      <h1 th:text="${driver.id == null} ? 'NEW DRIVER' : 'EDIT DRIVER'"></h1>
      <a href="/drivers" class="btn btn-ghost">← BACK</a>
    </div>

    <form class="form-card" action="#" th:action="@{/drivers/save}"
      ↪ th:object="${driver}" method="post">
      <input type="hidden" th:field="*{id}" />

      <div class="form-group">
        <label>Team *</label>
        <select th:field="*{team.id}" required>
          <option value="">- SELECT TEAM -</option>
          <option th:each="t : ${teams}"
            th:value="${t.id}"
            th:text="${t.name} + ' (' + ${t.series.name} + ')"></option>
        </select>

```

```
</div>
```

```
<div class="form-row">
```

```
  <div class="form-group">
```

```
    <label>First Name *</label>
```

```
    <input type="text" th:field="*{firstName}" required />
```

```
  </div>
```

```
  <div class="form-group">
```

```
    <label>Last Name *</label>
```

```
    <input type="text" th:field="*{lastName}" required />
```

```
  </div>
```

```
</div>
```

```
<div class="form-row">
```

```
  <div class="form-group">
```

```
    <label>Racing Number * (1-99)</label>
```

```
    <input type="number" th:field="*{racingNumber}" min="1" max="99" required  
      ↪ />
```

```
  </div>
```

```
  <div class="form-group">
```

```
    <label>Points</label>
```

```
    <input type="number" th:field="*{points}" required />
```

```
  </div>
```

```
</div>
```

```
<div class="form-actions">
```

```
  <button type="submit" class="btn btn-save">SAVE DRIVER</button>
```

```
  <a href="/drivers" class="btn btn-ghost">CANCEL</a>
```

```
</div>
```

```
</form>
```

```
</main>
```

```
</div>
```

```
<div class="bottombar">
```

```
  <div class="stat">
```

```
<span class="stat-dot"></span>
<span class="stat-label-bot">DB</span>
<span class="stat-value-bot">LIVE</span>
</div>
<div class="stat">
  <span class="stat-label-bot">Mode</span>
  <span class="stat-value-bot">EDITOR</span>
</div>
<div class="db">Java / Spring Boot 2.6 / Thymeleaf</div>
</div>

</body>
</html>
```